

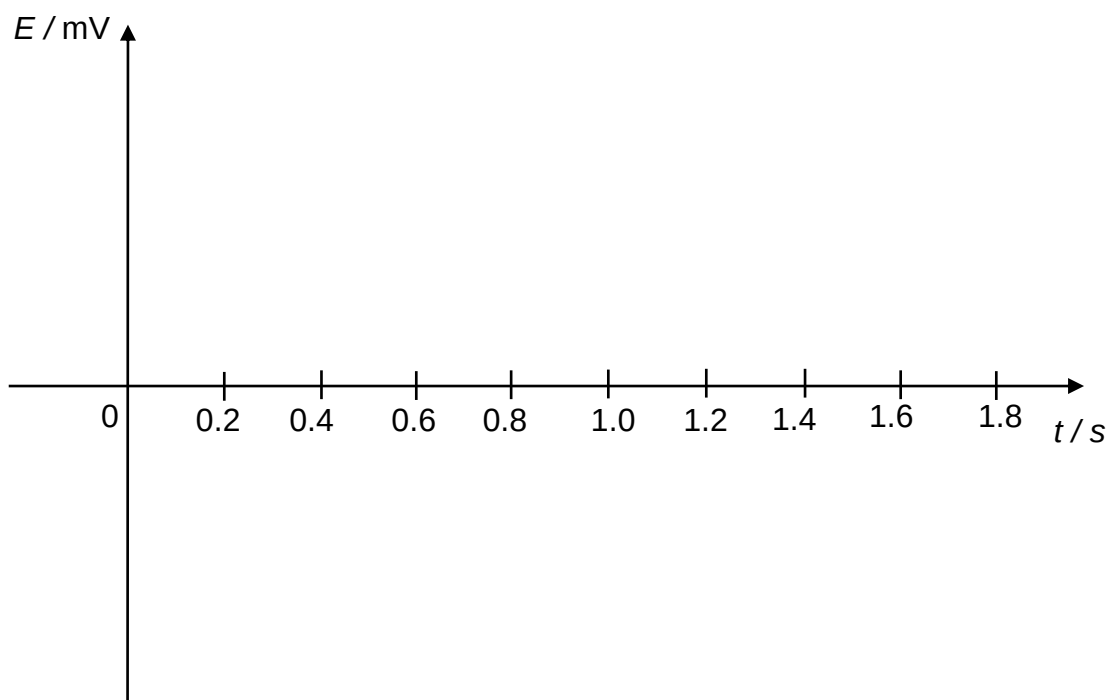
5	(a)	State Faraday's law of electromagnetic induction. 	[2]
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$$\begin{aligned}
 \text{Induced e.m.f.} &= \left| \frac{\Delta(N\phi)}{\Delta t} \right| \\
 &= \frac{(12 \times 10^{-4})}{0.30} \\
 &= 4.0 \times 10^{-3} \text{ V}
 \end{aligned}$$

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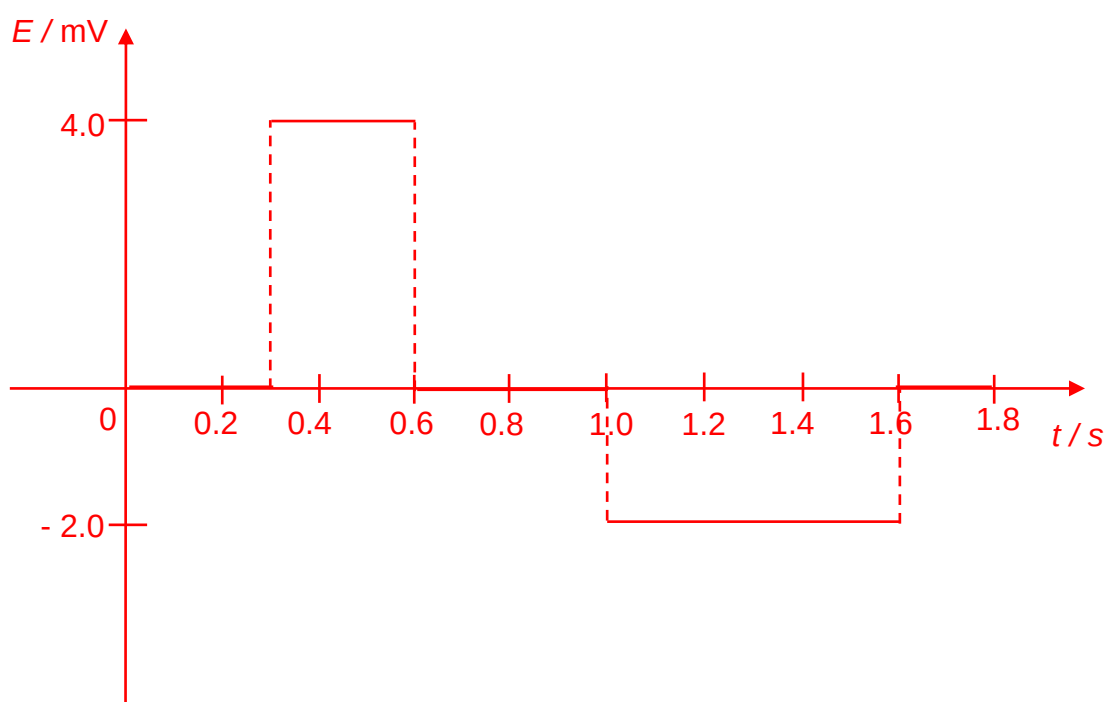
- (d) Use your answer to (c) to sketch on Fig. 5.3, the variation with time t of the e.m.f. E induced in coil C.



[4]

Fig. 5.3

- (d) **Solution:**



Sketch: $E = 0$ for $t = 0 \rightarrow 0.3 \text{ s}$, $0.6 \text{ s} \rightarrow 1.0 \text{ s}$, $1.6 \text{ s} \rightarrow 1.8 \text{ s}$

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		$E = 4 \text{ mV}$ (horizontal graph) – allow for ecf from previous part $E = 2 \text{ mV}$ (horizontal graph) Both being of opposite polarity to each other	1 1 1
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6	An ideal transformer is shown in Fig. 6.1	
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